

Figure 13: Portal of PHP-ML for machine learning using PHP

PHP based machine learning on the OpenShift PaaS

Machine learning is a powerful science that makes use of soft computing and meta-heuristic approaches for effectual predictive mining even from a huge dataset *(https://php-ml.readthedocs.io/en/latest/).* It has been traditionally used for fraud detection, market analytics, email spam filtering, malware analysis, fingerprint evaluation, face detection and many other applications. In machine learning, the algorithms are implemented in a way in which better classifications and predictions can be made—somewhat similar to the intelligence of the human brain. In traditional implementations, artificial neural networks are used with machine learning to solve complex classification problems.

A number of libraries and frameworks are available under Free and Open Source Software (FOSS) distribution for machine learning. FOSS libraries that can be integrated for research and development include PHP-ML, Apache Mahout, Shogun, Apache Singa, Apache Spark Mlib, TensorFlow, Oryx2, Accord.NET, Amazon Machine Learning, Scikit-Learn, H2O, ConvNetJS, etc.

PHP-ML is a powerful machine learning library used for R&D in the domain of machine learning for different applications. It integrates assorted algorithms in the form of classes and methods for high performance computing with the analytics from real-time datasets. PHP-ML has a rich set of algorithms implemented in PHP Scripts, and these can be easily integrated on the real-time cloud of OpenShift by uploading the code and mapping with GIT.

Key features and algorithms in PHP-ML

Association rule learning: Apriori
Classification: KNearestNeighbors, Naive Bayes, SVC, etc
Clustering: k-Means, DBSCAN
Cross validation: Random split, stratified random split
Feature extraction: Token count vectoriser, Tf-idf transformer
Metric: Accuracy, confusion matrix, classification report
Models management: Persistency
Math: Distance, matrix, set, statistic
Neural network: Multi-layer perceptron classifier
Preprocessing: Normalisation, imputation missing values
Regression: Least squares, SVR
Workflow: Pipeline

Datasets: Array, CSV, files, datasets for research: Iris, Wine and Glass

KNearestNeighbors Classifier in PHP-ML

KNearestNeighbors implements the k-nearest neighbours (k-NN) algorithm for solving the classification problems for a specific set of data items.

In the following example, inputs with their corresponding targets are specified in terms of classes '0' or '1'. If these values are carefully analysed, the corresponding classes can be mapped. In the dataset of [2, 5], [3, 6], [4, 7], the values in each set are in increasing order and there is a difference of +3, which is assigned to the class '0'. Similarly, in [4, 2], [5, 3], [7, 5], the values in each set are decreasing and assigned class '1'. This data can be trained using k-NN with the implementation of the *train()* function.

```
$input = [[2, 5], [3, 6], [4, 7], [4, 2], [5, 3], [7, 5]];
$target = ['0', '0', '0', '1', '1', '1'];
$classifier = new KNearestNeighbors();
$classifier->train($input, $target);
```

For the prediction of new input data, the *predict()* function is implemented. As in the following example, predict ([5, 7]) is passed as input, the output will be returned as class '0' because the values in [5, 7] are in increasing order and almost of the same behaviour as class '0'. The exact difference of +3 is not mandatory, because machine learning approaches make use of results with a higher degree of approximation, probability and optimisation.

```
$classifier->predict([[10, 6], [1, 3]]);
// The function will return ['1', '0'] depending upon the
pattern and behavior of input
```

Scope of R&D

As there are many applications for which classification and predictive mining can be used, the free and open source libraries can be integrated on the real-time clouds of Red Hat OpenShift, IBM Bluemix, Amazon, Google Apps Engine and many others, depending upon the algorithms to be used. The aspects and logs associated with performance, complexity, security and integrity can be analysed with the implementation of algorithms on real-time clouds. END

By: Dr Gaurav Kumar

The author is the MD of Magma Research and Consultancy, Ambala. He is associated with various academic and research institutes, where he delivers lectures and conducts technical workshops on the latest technologies and tools. You can contact him at *kumargaurav.in@gmail.com*. Website: *www.gauravkumarindia.com*.